



Original Article

Sexually attractive traits predict predation-threat sensitivity of male alternative mating tactics

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Although visual sexual signals, such as ornamental colors and courtship displays, and large body size in males are attractive to females in numerous species, they may also inadvertently attract the attention of eavesdropping predators and thus may be costly in terms of increasing individual risk of mortality to predation. Theoretically, more color ornamented and larger males should be more predation threat sensitive and suppress their sexual signaling and/or mating effort relatively more than their less color ornamented and smaller counterparts when under predation hazard. Here, we experimentally tested this hypothesis by quantifying concurrently the rates of alternative mating tactics (courtship displays, sneak mating attempts) expressed by male Trinidadian guppies (*Poecilia reticulata*) varying in color ornamentation and body size under a staged immediate threat of predation. Males suppressed their overall mating effort in response to the perceived predation threat, decreasing the frequency of their (presumably more conspicuous) courtship displays significantly more on average than the frequency of their sneak mating behavior. Statistically controlling for body length, more color-ornamented males were more threat sensitive in their courtship displays, but not sneak mating attempts, under predation hazard than drabber males. Controlling for body coloration, larger males exhibited lower courtship and sneak mating efforts than smaller males in both predation treatments, but body length only influenced threat sensitivity in sneak mating behavior. These results are consistent with both the threat sensitive hypothesis and asset protection principle and highlight the phenotype dependency and adaptive plasticity of alternative mating tactics in male guppies under varying predation risk.

Key words: body length, color ornamentation, guppy, mating tactics, predation risk, threat sensitivity.

INTRODUCTION

Sexual selection favors the evolution of secondary sexual traits in animals, such as extravagant ornamentation, exaggerated courtship displays, and large body size that increase their individual mating and reproductive success (Andersson 1994; Blanckenhorn 2000). In numerous species, highly color ornamented, vigorously courting, and/or larger males are generally more conspicuous (salient) and attractive to females, and thus enjoy greater mating and reproductive success on average, compared with drabber, less vigorously courting, and/or smaller male conspecifics (Andersson 1994; Houde 1997; Blanckenhorn 2000; Jennions et al. 2001; Hill 2006). However, visual sexual signals and large body size may also concurrently and inadvertently increase a male's conspicuousness and attractiveness to eavesdropping predators and other natural

enemies (Magnhagen 1991; Endler 1992; Zuk and Kolluru 1998; Blanckenhorn 2000; Bernal and Page 2023). Therefore, the evolution and maintenance of such traits should depend in theory on the balance or trade-off between their associated fitness benefits and costs; that is, on the balance between sexual and natural selection, respectively (Lima and Dill 1990; Magnhagen 1991; Andersson 1994; Sih 1994; Endler 1995; Engqvist et al. 2015; Lerch and Servedio 2023).

A recent meta-analysis of relevant literature (White et al. 2022) revealed that eavesdropping enemies exhibit moderate to strong preferences for individuals bearing conspicuous sexual signals compared to controls. As a consequence, individual animals bearing sexual signals suffer a significantly greater eavesdropping risk in the wild compared to conspecifics that do not exhibit such signals (White et al. 2022). However, the magnitude of this cost varies among signal modalities, with olfactory and auditory sexual signals increasing the risk of eavesdropping to a greater extent than visual

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sexual signals (White et al. 2022). Interestingly, this meta-analysis found no evidence overall for an effect of visual sexual signaling on the risk of predation in the wild, but the direction and size of effects were highly variable among studies. The observed on-average absence of additional marginal predation risk associated with visual sexual signaling (White et al. 2022) is potentially attributable to the general state-dependent nature of sexual signaling (Dougherty 2021), neophobia and/or dietary conservatism in predators that are exposed only infrequently to prey sexual signals or to sexual signals that are novel to them (Ord et al. 2021; White et al. 2022), and/or evolved plastic and fixed anti-eavesdropper strategies in the sexual signaling of prey species, such as signal pausing or cessation, reduction in signaling rate and altering signal structure, which reduce their detectability, localizability, or conspicuousness (Zuk and Kolluru 1998; Bernal and Page 2023).

Notwithstanding the latter meta-analytic general finding of White et al. (2022), there are, however, some empirical studies demonstrating that males expressing high levels of sexually selected color ornamentation are more conspicuous and attractive to visually guided predators, and as a result incur a greater marginal risk of mortality to predation, than less color ornamented male or non-ornamented female conspecifics (e.g., Godin and McDonough 2003; Stuart-Fox et al. 2003; Husak et al. 2006; Møller and Nielsen 2006; Modarressie et al. 2013; Kusche and Meyer 2014; Heinen-Kay et al. 2015; Marshall et al. 2015; Johnson and Candolin 2017). Similarly, sexually selected large body size in males can be associated with viability costs, including heightened mortality risk to predation (Blanckenhorn 2000). There is some empirical evidence for predators preferentially being attracted to and consuming larger males (e.g., Winemillar et al. 1990; Trexler et al. 1994; Blanckenhorn 2000; Quinn and Buck 2001; Rosenthal et al. 2001) or males possessing larger sexual ornaments (e.g., Rosenthal et al. 2001; Hernandez-Jimenez and Rios-Cardenas 2012) in prey species wherein females are also preferentially attracted to males with larger body size or larger sexual ornaments.

The individual risk of predation that is potentially associated with conspicuous color ornamentation and/or large body size may be further elevated for a courting individual if courtship display behavior per se attracts the attention of predators, amplifies the effect of coloration or body size on predator attraction, and/or reduces the courter's antipredator vigilance and predator avoidance (Lima and Dill 1990; Magnhagen 1991; Sih 1994; Zuk and Kolluru 1998; Dukas 2004; Clark et al. 2016; Johnson and Candolin 2017). Whenever this is the case, the most color-ornamented and/or larger males exhibiting the highest courtship efforts in the population would likely be the ones most vulnerable to predation, all else being equal (cf. Reynolds 1993). If so, then one would expect selection to favor those males who facultatively reduce in the short term their courtship effort and/or the conspicuousness of their body coloration in proportion to their perceived level of predation risk (i.e., exhibiting threat sensitivity, *sensu* Helfman 1989), but at the potential cost of lost mating opportunities (Zuk and Kolluru 1998), in a manner that maximizes their lifetime reproductive success (Lima and Dill 1990; Magnhagen 1991; Reynolds 1993; Sih 1994; Engqvist et al. 2015). Such individual plasticity in sexual signaling is common and appears adaptive as an anti-eavesdropper and antipredator strategy (Lima and Dill 1990; Sih 1994; Zuk and Kolluru 1998; Bernal and Page 2023).

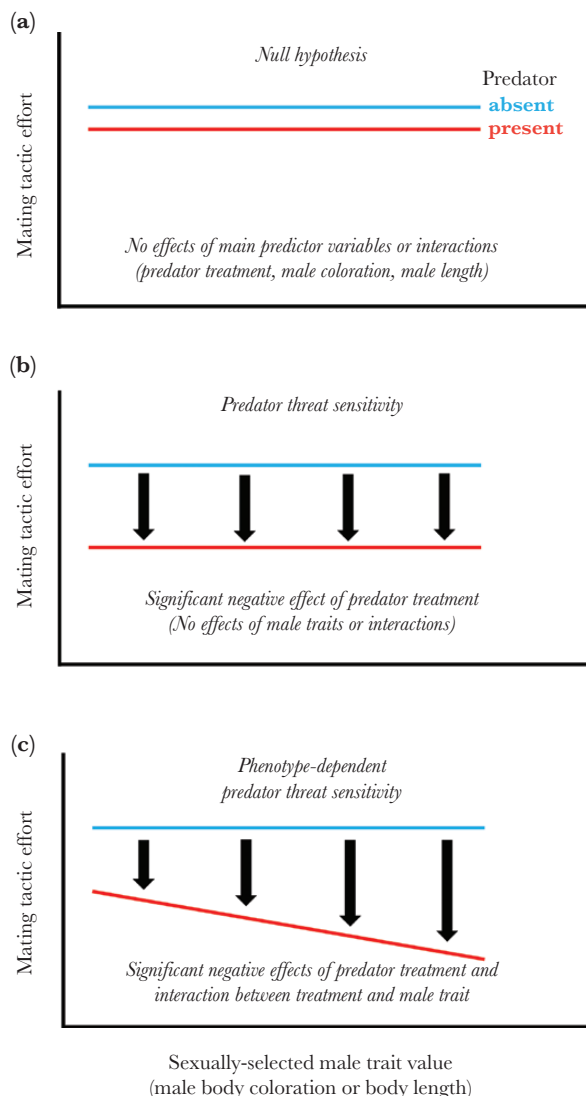
The extent to which an individual male reduces his visual sexual signaling under threat of predation (i.e., its threat sensitivity) depends on a number of factors, including the magnitude of his *direct*

perception of the predation threat, the expected mating payoff of signaling, his vulnerability to predation and his current condition (i.e., ability to bear the associated cost of signaling) (Lima and Dill 1990; Reynolds 1993; Sih 1994; Zeh and Zeh 1988; Zuk and Kolluru 1998; Jennions et al. 2001; Lindström et al. 2006; Dougherty 2021; Lerch and Servodio 2023). Alternatively, but not mutually exclusive, threat-sensitivity in sexual signaling behavior by individual males under acute predation risk may be mediated *indirectly* by observed changes in the general activity (Lima and Dill 1990) and antipredator behavior of nearby females, such as increased shoaling and predator "inspection" (Magurran and Nowak 1991; Dill et al. 1999; Evans et al. 2002), their reduced sexual activity and weakened mating preferences (Godin and Briggs 1996) in response to perceived danger, and/or their increased mating preference for apparently bolder males over timid ones (Godin and Dugatkin 1996), which in turn could influence male mating effort, under the threat of predation.

Alternative male mating strategies and tactics within natural populations are ubiquitous (reviewed in Brockmann 1991; Taborsky 2008) and their expression and relative frequencies can be influenced by ecological factors such as local predation risk for example (e.g., Luyten and Liley 1985; Magurran and Seghers 1994; Godin 1995). As such, for species that exhibit alternative male mating tactics (e.g., courtship displays vs. coercive sneak mating attempts), any predator-mediated relative reduction in individual courtship effort should be greater on average than that for sneak mating effort, because bouts of the former tactic are generally longer and potentially more visually salient to predators than those of the latter (e.g., Luyten and Liley 1985; Houde 1997; Pilastro et al. 2007).

A recent evolutionary model of optimal courtship risk taking (Engqvist et al. 2015) predicts that, all else being equal, sexually attractive males in a population should be more risk averse and suppress their courtship display rate to a greater extent than relatively unattractive males when their perceived risk of mortality to predation has increased, assuming that more sexually attractive males incur higher predation costs when displaying (Magnhagen 1991; Zuk and Kolluru 1998; Clark et al. 2016; Johnson and Candolin 2017) and have more to lose in terms of "assets" (residual lifetime reproductive value) according to the asset protection principle (Clark 1994). Other than for a few studies (Reynolds 1993; Reynolds et al. 1993; Candolin 1998; Kraak et al. 2000; Warner and Dill 2000; Fowler-Finn and Hebets 2011; McQueen et al. 2017), little is known about the relationship between the degree of threat sensitivity of mating effort in reproductively active males and their sexual attractiveness to females, which is of particular interest in our current study.

Here, using the Trinidadian guppy (*Poecilia reticulata*) as a model study system, we experimentally tested the above-mentioned theoretical predictions that, within a population, (1) males should suppress their individual mating effort (courtship displays and sneak mating attempts) when under perceived acute predation threat (Figure 1B), (2) any predator-mediated reduction in courtship rate in males should be greater on average than any predator-mediated reduction in sneak mating attempts, and (3) the magnitude of the suppression of mating effort under predation threat (i.e., degree of threat sensitivity) in individual males should be conditionally proportional to their sexual attractiveness, as measured by body color ornamentation and body size, respectively (Figure 1C). Our null hypothesis is that individual males, irrespective of their phenotype, do not adjust their mating effort when they perceive their immediate risk of predation to have increased (Figure 1A). The guppy

**Figure 1**

Hypothetical relationships between male mating effort (courtship display rate or sneak mating attempt rate) under the presence or absence of a predator and male sexually selected phenotypic traits (body coloration or body length), and their interactions. A male's perception of his immediate risk of predation is assumed to be positively related to his sexual attractiveness to females, as measured by his body coloration and body length. (a) Null hypothesis; no statistically significant effects of predator treatment interactions between the fixed predictor variables. (b) Predator threat sensitivity in male mating tactic effort; significant negative effect of predator treatment, but no effects of male traits or interaction between predator treatment and male traits. (c) Phenotype-dependent predator threat sensitivity in male mating tactic effort; significant negative effect of predator treatment and interaction between predator treatment and male traits.

is a highly suitable model species for the current study for reasons summarized below.

MATERIALS AND METHODS

Model study system

The Trinidadian guppy is a small, sexually dimorphic freshwater fish species endemic to Trinidad, northeastern Venezuela and

surrounding continental islands (Magurran 2005). Adult males are highly polymorphic for body coloration, whereas adult females lack ornamental coloration and are uniformly drab olive/beige (Houde 1997; Magurran 2005). The body coloration pattern of individual males is unique and comprises of spots/patches of pigmented and structural iridescent colors (e.g., Endler 1978, 1980, 1991; Kemp et al. 2008; see Supplementary Figure S1). Male body coloration in the guppy is heritable and responsive to selection (reviewed in Brooks and Postma 2011). Carotenoid-based pigmented colors (i.e., red, orange, yellow) in this species are reliable indicators of individual male vigor/condition (Nicoletto 1991; Houde and Torio 1992; Godin and Dugatkin 1996; Kodric-Brown and Nicoletto 2005), functional fertility (Pitcher and Evans 2001; Pilastro et al. 2002; Locatello et al. 2006; Pitcher et al. 2007; Evans 2010), sexual attractiveness (Grether 2000; Brooks and Endler 2001a; b), mating and reproductive success (Houde 1997; Brooks and Endler 2001a; Evans et al. 2003; Becher and Magurran 2004; Pilastro et al. 2004; Devigili et al. 2015; Gordon et al. 2015; Herdegen-Radwan et al. 2021), offspring viability (Evans et al. 2004), and overall fitness in the wild (Gordon et al. 2015). Wild adult male guppies also vary in body size within and among populations (e.g., Reznick et al. 1990; Hendry et al. 2006; Auge et al. 2016). Body size is also heritable and responsive to selection in this species (Reznick and Bryga 1987; Reznick et al. 1990; Reynolds and Gross 1992). The body length of individual male guppies predicts their functional fertility (Pitcher and Evans 2001; Pilastro et al. 2002), individual mating and reproductive success (Becher and Magurran 2004; Magellan et al. 2005; Devigili et al. 2015; Glavaschi et al. 2022), and the quality (i.e., growth and fecundity) of their offspring (Reynolds and Gross 1992).

Depending on the population, females prefer on average the more color-ornamented and/or larger of available males as mates (e.g., Reynolds and Gross 1992; Endler and Houde 1995; Brooks and Endler 2001a; b; Dosen and Montgomerie 2004; Herdman et al. 2004; Magellan et al. 2005; Auld et al. 2016). However, male guppies possessing visually conspicuous ornamental coloration may incur the cost of a greater individual risk of mortality to predation than drabber conspecifics (e.g., Haskins et al. 1961; Endler 1978, 1980; Godin and McDonough 2003; Weese et al. 2010), a survival cost that can vary depending on population (Weese et al. 2010), the particular class of color (Endler 1978; Gordon et al. 2015), relative frequencies of color morphs in the population (Olendorf et al. 2006), and local habitat features (Endler 1995; Millar et al. 2006), for example. Fish predators of the guppy can be size-selective in their prey choice and preferentially predate larger guppies over smaller ones (Liley and Seghers 1975; Endler 1980; Reznick et al. 1990; Winemillar et al. 1990; Johansson et al. 2004), depending on habitat and population (e.g., Mattingly and Butler 1994). Consistent with selection by visually guided predators against conspicuous body coloration patterns and larger body size, adult male guppies in natural populations that experience high predation intensity are generally less color ornamented and smaller on average than males in populations experiencing weaker predation intensity (Endler 1978, 1980, 1995; Winemillar et al. 1990; Millar et al. 2006; Reznick and Endler 1982; Reznick and Bryga 1987; Reznick et al. 1990, 1996; Gordon et al. 2015; but see Heckley et al. 2022).

The guppy is also a live-bearing species with internal fertilization. It exhibits a non-resource-based, promiscuous mating system and occurs in mixed-sex shoals in nature (Houde 1997; Magurran 2005). Males exhibit two alternative mating tactics. Individuals may facultatively either court a female prior to attempting a consensual copulation with her or attempt to sneak (coerced) copulate

with a female without first displaying to her or prior to a receptive response from her (Liley 1966; Houde 1997). A courtship display typically consists of a male positioning himself 2–10 cm in front or to the side of a female, arching its body into a sigmoid S-shape with open or closed dorsal and caudal fins, and quivering its body vertically for several seconds, commonly 2–15 s (Liley 1966; Endler et al. 2022; see Supplementary Videos S1 and S2). A sneak mating attempt consists of a male stealthily approaching a non-responsive female from behind, below or to one side and then rapidly darting toward her to attempt a sneak copulation by thrusting its gonopodium (intromittant organ) toward her gonopore (i.e., a gonopodial thrust; *sensu* Liley 1966; see Supplementary Video S3); a sneak copulation attempt is very brief (typically ≤ 1 s).

Female guppies tend to be preferentially attracted to males who court most vigorously (Kodric-Brown 1993; Godin and Briggs 1996; Kodric-Brown and Nicoletto 2001; Kolluru et al. 2009). Courtship display rate can indicate male health/condition (Kennedy et al. 1987; Abrahams 1993; Kolluru et al. 2009), functional fertility (Matthews et al. 1997; Pitcher and Evans 2001), and reproductive success (Evans and Magurran 2001; Devigili et al. 2015; Glavaschi et al. 2022). Conversely, females tend to avoid males who attempt to copulate with them coercively (Magurran and Nowak 1991; Magurran and Seghers 1994; Darden and Croft 2008). Although courtship is on average the more successful of these two mating tactics in the guppy (Pilastro and Bisazza 1999; Pilastro et al. 2007), it is likely the more costly in terms of greater energetic expenditure and increased associated risk of predation because courtship displays and courtship copulations are significantly longer in duration, and presumably more visually conspicuous (more salient) to predators, than sneak mating attempts and sneak copulations (Liley 1966; Luyten and Liley 1985; Houde 1997; Pilastro et al. 2007).

Given the aforementioned fitness-related attributes of color ornamentation, large body size and courtship in male guppies, it is reasonable to assume that more color-ornamented, larger and/or courting males are more visually conspicuous and have more “assets” (residual lifetime reproductive value) to protect (*sensu* Clark 1994), and should therefore be more predation threat sensitive and risk averse, than their less ornamented, smaller, and/or non-courting counterparts.

Notwithstanding the above-mentioned empirical evidence for positive relationships between male body size and (1) visual conspicuousness and individual risk of mortality to predation and (2) sexual attractiveness and mating success in the guppy, there are other potential mechanisms through which body size may influence individual predation risk and the success of alternative mating strategies. For example, in teleost fishes, larger individuals have higher burst (fast start) and sustained swimming speeds in general (Blake 1983), and may thus be less vulnerable to predation, than smaller males (Godin 1997). Larger individuals may also be less susceptible to predation owing to gape limitations in their natural fish predators (Godin 1997), but this is not likely given the small body lengths of adult male guppies relative to the body lengths of the larger pike cichlid predators used in the current study (range of length ratios = 1:4.08–1:7.25; cf. Godin 1997); these pike cichlids could readily handle and ingest captured male guppies when given the opportunity. In addition to their greater sexual attractiveness, larger male guppies might also have greater success at sneak mating owing to their potential higher burst swimming speeds (Odell et al. 2003), allowing them to more quickly close the distance to the typically faster female, than their smaller counterparts. To our knowledge, there is no direct evidence for this during sneak mating attempts

in male guppies. Conversely, smaller males might be less conspicuous when approaching an unreceptive female and more maneuverable when inserting the gonopodium into the female's gonopore than larger males, allowing them to more successfully inseminate a female by sneak mating. This is generally the case in poeciliid fishes, including the guppy (Bisazza and Pilastro 1997; Pilastro et al. 1997).

Subjects and laboratory husbandry

As experimental subjects, we used wild adult male and female guppies that were haphazardly collected under permit with a hand-held seine in the main Quaré River (10°40'N, 61°12'W) in Trinidad. This population is subjected to relatively high fish predation pressure (Deacon et al. 2018). Pike cichlids (*Crenicichla frenata*), a major natural predator of the guppy (Magurran 2005), were also collected from the Quaré River and used to experimentally stage an immediate threat of predation toward test guppies in the laboratory. We chose Quaré River guppies as subjects for our current study because guppies originating from high-predation populations in Trinidad generally express greater threat sensitivity in their antipredator (Botham et al. 2008; Brown et al. 2009) and mate-choice behavior (Godin and Briggs 1996) than those from low-predation populations.

Following their collection in the wild, the fish were placed in containers filled with aerated water and transported by air to our institutional aquatic facility. The guppies and pike cichlids were held in separate glass aquaria, out of view of each other. The cichlids were placed individually in separate aquaria, whereas the guppies were kept in large mixed-sex aquaria. Each aquarium contained continuously filtered aged well water (24–26 °C), a gravel substratum, clay pots, and plastic plants anchored into the bottom for environmental enrichment and potential refuge, and was exposed to a 12 h L:12 h D illumination cycle provided by an overhead Sun-Glo™ fluorescent tube (R.C. Hagen Inc., Montréal, Canada) which simulates the energy spectrum of sunlight. Guppies were fed *ad libitum* commercial flake food (NutraFin™, R.C. Hagen Inc.) twice daily, supplemented with live brine shrimp nauplii (*Artemia franciscana*), whereas the cichlid predators ($N = 10$, 12–14 cm total length) were fed freeze-dried euphausiid shrimps, *Euphausia pacifica* (Murex Ocean Plankton™, R.C. Hagen Inc.) twice daily and occasionally guppies (of unknown provenance) obtained from a commercial aquarium fish store. The fish were acclimatized to these husbandry conditions for at least 1 month before being used in the current study. We only used guppies that were free of visible ectoparasites as experimental subjects, because ectoparasites can adversely affect male body coloration (Houde and Torio 1992) and courtship behavior (Kennedy et al. 1987; Kolluru et al. 2009) and female sexual responsiveness and mate choice (López 1999) in this species.

Experimental apparatus

The experimental apparatus consisted of two identical glass aquaria (60 × 30 × 30 cm, L × W × H), each containing a gravel substratum and continuing filtered and aerated well water (25 cm depth) maintained at 24–26 °C. Each aquarium was divided into two unequal compartments by two adjacent Plexiglas partitions (1 mm apart), one clear and stationary (sealed with silicone to the inside of the aquarium) and the other opaque and removable. The larger of the compartments (45 × 30 × 30 cm, L × W × H) contained one focal male and two stimulus female guppies. The smaller compartment (15 × 30 × 30 cm) housed a live pike cichlid predator in one of the paired test aquaria (i.e., the experimental aquarium),

whereas it was left empty in the other (control) aquarium. Both aquaria were covered externally on three sides with tan paper and with the fourth side left open for observations. The entire apparatus was located behind a blind and illuminated overhead with a Sun-Glo fluorescent tube ($10.0 \mu\text{E}/\text{m}^2/\text{s}$ at the surface). This ambient experimental light intensity falls within the range of lighting conditions experienced by guppies in their natural riverine habitats during early morning and late afternoon/early evening (Endler 1987; Godin 1995). All behavioral observations on the guppies were made undisturbed from behind the blind by one of us (H.E.M.) to avoid inter-observer measurement errors or bias and blind of individual male body color ornamentation scores and body length.

Experimental protocol

A typical trial consisted of paired 15-min treatments (i.e., predator absent vs. predator present), during which we quantified the frequencies of courtship displays and sneak copulation attempts that a focal male guppy directed toward two females from the same population. The sum of a male's courtship displays and sneak mating attempts during a given period constitutes its mating effort (cf. Godin and Auld 2013). For a given trial, one of the paired treatments (i.e., 15-min observation period) was conducted in early morning (08:00–09:30 h) and the other in late afternoon (16:00–17:30 h), times of the day when male guppies are known to sexually pursue females most often in nature (Endler 1987; Houde 1997). The stimulus female guppies were gravid (pregnant) adults chosen from a holding tank different from that of the test male to avoid prior individual familiarity. A sex ratio of 1 male: 2 females was used to reflect female-biased sex ratios common in wild guppy populations in Trinidad (Magurran 2005), including the Quaré River (Godin 1995). We importantly used gravid females, who are generally unresponsive to male courtship and copulation attempts (Liley 1966; Houde 1997), as stimulus fish because it is the most common reproductive state of adult female guppies in the wild (Houde 1997), to ensure that male mating behavior would not be confounded by female sexual responses to male sexual activity and to minimize variation in male behavior caused by any differences in female reproductive state. Although male guppies generally prefer unmated over mated (gravid) females as mates when available (Guevara-Fiore et al. 2009, 2010) and most adult female guppies in natural populations in Trinidad are pregnant at any given time (Houde 1997), males nonetheless sexually pursue, court and attempt to mate with previously-mated gravid females in both the wild and laboratory (Houde 1997; Guevara-Fiore et al. 2010; Jeswiet et al. 2011; Godin and Auld 2013).

Prior to the onset of a trial in early morning, a test male guppy and two stimulus females were removed from their respective holding aquaria and placed into the larger observation compartment of each of the two test aquaria, with the opaque partition in place to obstruct the view of the adjacent predator compartment. A pike cichlid predator was then haphazardly selected from among the 10 available individuals and transferred into the predator compartment of the experimental aquarium designated for the predator-present treatment. The fish were allowed 45–60 min to acclimatize to the aquaria before the onset of the morning observation period. Following this acclimatization period, the opaque partition was raised remotely and slowly using a string and pulley system, and the guppies were allowed to view into the predator compartment, which contained a live pike cichlid in the predator-present treatment but was empty in the predator-absent treatment. Once the guppies resumed swimming, we then recorded on

a voice-activated tape recorder the real-time occurrences of sigmoid courtship displays and sneak mating attempts (gonopodial thrusts) directed by the test male toward the females for 15 min. At the end of this morning observation period, the opaque partition in the experimental aquarium was lowered, and the male and female guppies from the experimental (predator-present) aquarium were gently dipnetted and transferred to the control (predator-absent) aquarium, or vice versa, and allowed to acclimatize before re-testing. In late afternoon of the same day, the protocol described above was followed and the 15-min observation period was repeated on the same guppies for the alternate paired treatment. Therefore, each male guppy subject was tested in a matched pair of treatments (predator absent vs. predator present), with their order of presentation alternated between trials on the same day across the subjects. At the end of a test day, male and female guppies were removed from the test aquaria, placed into separate holding aquaria, and not re-used. The cichlid predator was returned to its home holding aquarium and later re-used in subsequent trials as necessary. We then replaced the water in the two test aquaria with fresh water in preparation for the next test day. A total of 53 male guppy subjects were similarly tested. The audio tapes for individual trials were numbered and the frequency of male mating tactics later transcribed from them blind of predation treatment and male phenotype.

Measurements of body color ornamentation and body length

Following their use in above experiment, the body color ornamentation and body length of each male subject were respectively quantified and measured digitally as follows. To avoid the use of anesthetics and to minimize handling stress, both of which can potentially distort coloration patterns and color intensity in teleost fishes (e.g., Gray et al. 2011), males were individually placed in a clear and perforated rectangular Plexiglas canister ($29 \times 7 \times 8 \text{ mm}$, $L \times W \times H$, inside dimensions) underwater, which was attached to a clear plastic rod and mounted to an overhead clear Plexiglas plate. A small metric ruler was permanently affixed to the bottom outside edge of this canister for scale (Supplementary Figure S1). Because the small dimensions of the canister closely approximated the body length and depth of male guppies, the potential for movement was very limited. This canister, containing a stationary male guppy, was then submerged horizontally into a larger clear Plexiglas box filled with aged well water ($24\text{--}26^\circ\text{C}$) and covered at the back with tan paper (same paper as used on the experimental aquarium). The tan paper provided a standardized uniform background against which to photograph the fish. The canister was illuminated overhead with a Sun-Glo fluorescent tube. Under these standardized lighting and background conditions, the right side of the stationary male was photographed on Kodachrome 100 ASA film using a 35-mm Olympus OM 2 camera (fitted with a 105 mm macro lens and an auto shoemount flash set to F4.0/8.0) mounted on a tripod placed 30 cm from the fish. The entire procedure typically took less than 2 min. The color photographs were digitized, saved in .jpg format (5245×6787 pixels, 617 dpi) and analyzed using the open-source software *ImageJ* (<http://rsbweb.nih.gov/ij/>; Schneider et al. 2012).

The body coloration pattern of individual male guppies is comprised of spots or patches of pigmented and structural iridescent colors that vary in location, size, hue, chroma, reflectance, and contrast (Endler 1978, 1991; Kemp et al. 2008). Here, our photographic and analytical equipment only allowed us to quantify the number and size of a male's colored spots and

patches. With *ImageJ*'s tracing tool, we separately quantified the area (in mm²) of carotenoid- and pteridine-based spots/patches of red-orange-yellow (hereafter orange color, *OR*), discrete melanin black (*BL*) and less discrete melanin “fuzzy black” (*FBL*) pigmentation and iridescent (*IRID*) structural coloration (silver, blue, green) on the right side of each male's body, including the caudal fin; see [Supplementary Figure S1](#)). We also quantified the male's lateral body area as represented by its right-side profile (excluding dorsal and ventral fins). The number, location, and areal coverage of color spots/patches for each of the above color classes are strongly positively correlated across the right and left sides of individual male Trinidadian guppies ([Kemp et al. 2008](#)). Area measurements were taken twice and the average values calculated from the repeated measurements. The total body length (from the tip of the snout to the end of the caudal fin) of each male was also repeatedly measured and averaged to the nearest 0.1 mm with *ImageJ*.

Only one of us (J.-G.J.G.) photographed the fish and analyzed all photographs to avoid any inter-observer measurement error or bias. All measurements were taken blind of male mating behavior. To control for inter-individual variation in body size, each separate area of coloration (*OR*, *BL*, *FBL*, *IRID*) was divided by the lateral body area (*BODY*) of the fish and thus represented separate univariate color elements. We then calculated the relative area of total body coloration ($[OR + BL + FBL + IRID] / BODY$) for each male as their individual composite color ornamentation score. On average ($\pm$ SE), male test guppies were 23.3 ± 0.3 mm (range = 19.3–29.4 mm) in total body length and $14.84 \pm 0.73\%$ (range = 6.50–26.27%) of their body's right side was covered with pigmented and iridescent colors.

Female guppies most likely assess and respond to overall or composite male color ornamentation and male relative body size when choosing mates ([Reynolds and Gross 1992](#); [Endler and Houde 1995](#); [Brooks and Endler 2001a, b](#); [Dosen and Montgomerie 2004](#); [Herdman et al. 2004](#); [Auld et al. 2016](#)). The visual conspicuousness (*sensu* [Endler 1990](#)) of the body coloration pattern of a prey depends on its overall visual contrast with the background against which it is viewed by its predators ([Endler 1978, 1990, 1991](#); [Endler et al. 2022](#)). Given that male guppies were photographed, and also viewed by the cichlid fish predators, under standardized lighting conditions and against a uniform tan background in the current study, we assumed here that the visual conspicuousness (saliency) of individual male guppies to their fish predators would be positively correlated with their respective sexual attractiveness to females, as measured by their color ornamentation score and body length. We believe that this is a reasonable assumption given that cichlid fish predators preferentially attack and consume male guppies that are more color ornamented over less ornamented ones ([Godin and McDonough 2003](#)) and large guppies over smaller ones ([Liley and Seghers 1975](#); [Mattingly and Butler 1994](#); [Johansson et al. 2004](#)) when given the choice under standardized laboratory conditions. We recognize, however, that this may not necessarily always be the case in the guppy's natural habitats in Trinidad, where ambient light conditions and underwater visual environments are more varied and complex ([Endler 1991, 1995](#); [Endler et al. 2022](#)) and spatiotemporal variance in viability selection exists ([Millar et al. 2006](#); [Weese et al. 2010](#); [Barbosa et al. 2018](#)), and given that co-occurring fish predators and guppies differ in their visual sensory physiology and perception of male colors and color contrasts ([Endler 1991](#); [Endler et al. 2022](#)).

Statistical analyses

All data were analyzed in R v4.1.3 ([R Core Team 2022](#)) and all statistical tests are two-tailed. We used the R packages “tidyverse” ([Wickham et al. 2019](#)), “readxl” ([Wickman and Bryan 2022](#)), “ggbeeswarm” ([Clarke et al. 2022](#)), and “patchwork” ([Petersen 2022](#)) for data wrangling and visualization. Moreover, we used the R packages “glmmTMB” ([Brooks et al. 2017](#)) for analyses, “DHARMA” for diagnostic checks ([Hartig 2021](#)), and “emmeans” ([Lenth 2022](#)) for extracting the coefficients of trend lines (along with their associated 95% confidence intervals [CIs] and contrast) for each predator treatment. For all models carried out, diagnostic checks included both visual inspection of patterns of residuals and simulation tests for zero-inflation and overdispersion. We used the Akaike information criterion (AIC) for model comparisons (with smaller AIC values preferred, and a threshold Δ AIC of 2 considered to represent a significant difference in model fit; [Burnham and Anderson 2002](#)).

First, we initially attempted to test with a single omnibus model our a priori prediction that male courtship displays and sneak mating attempts directed toward females would be differentially suppressed in response to a perceived immediate predation threat (cf. [Figure 1B](#) vs. [Figure 1A](#)). We fitted the observed frequencies of these two behaviors as a single response variable (Mating behavior rate) in a generalized linear mixed model (GLMM) using the *glmmTMB()* function in the R package “glmmTMB.” Mating tactic type (binary “dummy” variable, where 0 represents courtship displays and 1 represents sneak mating attempts) and predator treatment (binary “dummy” variable, where 0 represents predator absent and 1 represents predator present), and their interaction, were included in the model as the main fixed effect explanatory variables. We also fitted the order of predator treatment as a dummy variable—although in this case centered at 0 such that model estimates would be given at the average (i.e., where the first observation was given the value -0.5 and second observation 0.5). The inclusion of predator treatment order controls for differences in the order of presentation of the two predator treatments across subjects. Lastly, we fitted male identity (ID) as a random effect, but allowed this to both vary and covary across the two types of mating tactic. Male ID as a random effect accounts for repeated behavioral measures on individual focal males across the paired predator treatments, and estimates separate variances for each mating tactic type along with their correlations (where, e.g., a positive correlation would suggest males that have relatively high values for courtship also have relatively high values for sneak mating behaviors).

We first assumed a Poisson error distribution given that the behaviors are count data, but model checks showed some overdispersion and strong zero-inflation. Because negative binomial errors failed to significantly improve the fit, we alternatively opted for an omnibus zero-inflated Poisson model. Zero-inflated models, such as the latter one, are a class of models capable of dealing with excess zero counts and a variety of response distributions. They are two-component mixture models combining a point mass at zero (i.e., zero-inflated component) with an observed count distribution (i.e., count component) ([Zeileis et al. 2008](#)). The zero-inflated component describes the probability of observing an extra point mass of (structural) zeros in the response distribution ([Brooks et al. 2017](#)). These models also allow both fixed and random effects to be fitted in the zero-inflation and count components of the model separately. Predictor variables in the count component are as described above. Here, we allowed zero-inflation to vary by both mating tactic type and predator treatment, as well as by male ID (as a random effect).

We did not allow the effect of male ID to (co)vary across mating tactic types in the zero-inflation component as this prevented the model from converging. Visual checks and tests using DHARMA indicated that this model met the assumptions of regression and solved issues of both overdispersion and zero-inflation. For both the count and zero-inflated components of the model, the interaction between predator presence and mating tactic type tests our a priori prediction that the magnitude of any predator-mediated reduction in courtship rate in males should be greater on average than the suppression in sneak mating attempts (either via decreasing the count or increasing the likelihood of zeroes).

Second, we used separate GLMM models for each mating tactic to test our a priori prediction that the magnitude of predator threat sensitivity of courtship displays and sneak mating attempts in individual male guppies should be proportional to their visual conspicuousness and attractiveness (as measured by body color ornamentation and body length) to mates as well as to predators (cf. Figure 1C). For each mating tactic in turn, we fitted a model that included predator treatment (as described above), male color ornamentation score and log body length, along with all interactions, as fixed effects. We also included predator treatment order as a covariate. We mean-centered and scaled to units of 1 standard deviation (SD) male color ornamentation scores (proportion total color), scaled to SD units male body length (log mm), and mean-centered treatment order. Mean-centering and scaling place each predictor variable on the same scale, which facilitates model fitting and interpretation of results (Schielzeth 2010). Male ID was fitted as a random effect. After using this “full” model specification to generate a reasonable model fit (described below), we iteratively removed the three-way interaction and the interaction between male color ornamentation and log body length (while checking that these did not improve the model fit), as these interactions were not of major relevance to our biological hypotheses.

For courtship displays, neither a standard Poisson model nor a negative binomial one fitted the data well (i.e., evidence of zero-inflation and overdispersion), but a zero-inflated Poisson model did. In the zero-inflated component of the latter model, we tried fitting predator treatment and the male color ornamentation and body length variables as well as the random effect of male ID, but none of these improved the fit relative to a simple global effect of zero-inflation. For sneak mating attempts, we found that a negative binomial error distribution provided a good fit to the data.

For each of the above final GLMMs, we separately plotted the raw data and the predicted relationship (with its estimated slope $[\beta]$ and associated 95% CI) between the particular behavioral variable of interest (courtship display or sneak mating attempt rate) and either male color ornamentation or body length, while holding the other variables in the model constant. In the classical (frequentist) sense, if the 95% CI associated with the slope (β) of a predicted relationship overlapped zero, then we deemed this β coefficient to be statistically nonsignificant. For each of the four separate plots (see Figure 3), we tested the difference in the slopes ($\Delta\beta$) of the predicted relationships for the paired predator-absent and predator-present treatments, respectively, using the *emtrends()* function in the R package “emmeans” (Lenth 2022). The difference in paired slopes ($\Delta\beta$) being compared was considered a measure of the predator threat sensitivity of the particular fitted behavioral variable.

Ethical statement

Our study received prior approval (protocol #95-13) from the institutional Animal Care Committee at Mount Allison University where the experiment was carried out, and thus adheres to the guidelines for the care and use of animals in research of the Canadian Council on Animal Care, the Animal Behavior Society and the Association for the Study of Animal Behaviour, and the laws of Canada. In particular, we took appropriate measures to

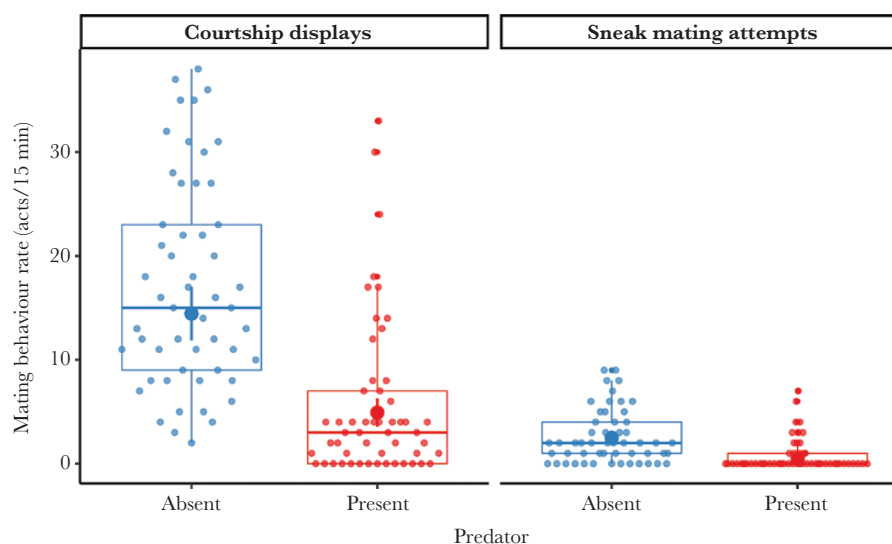
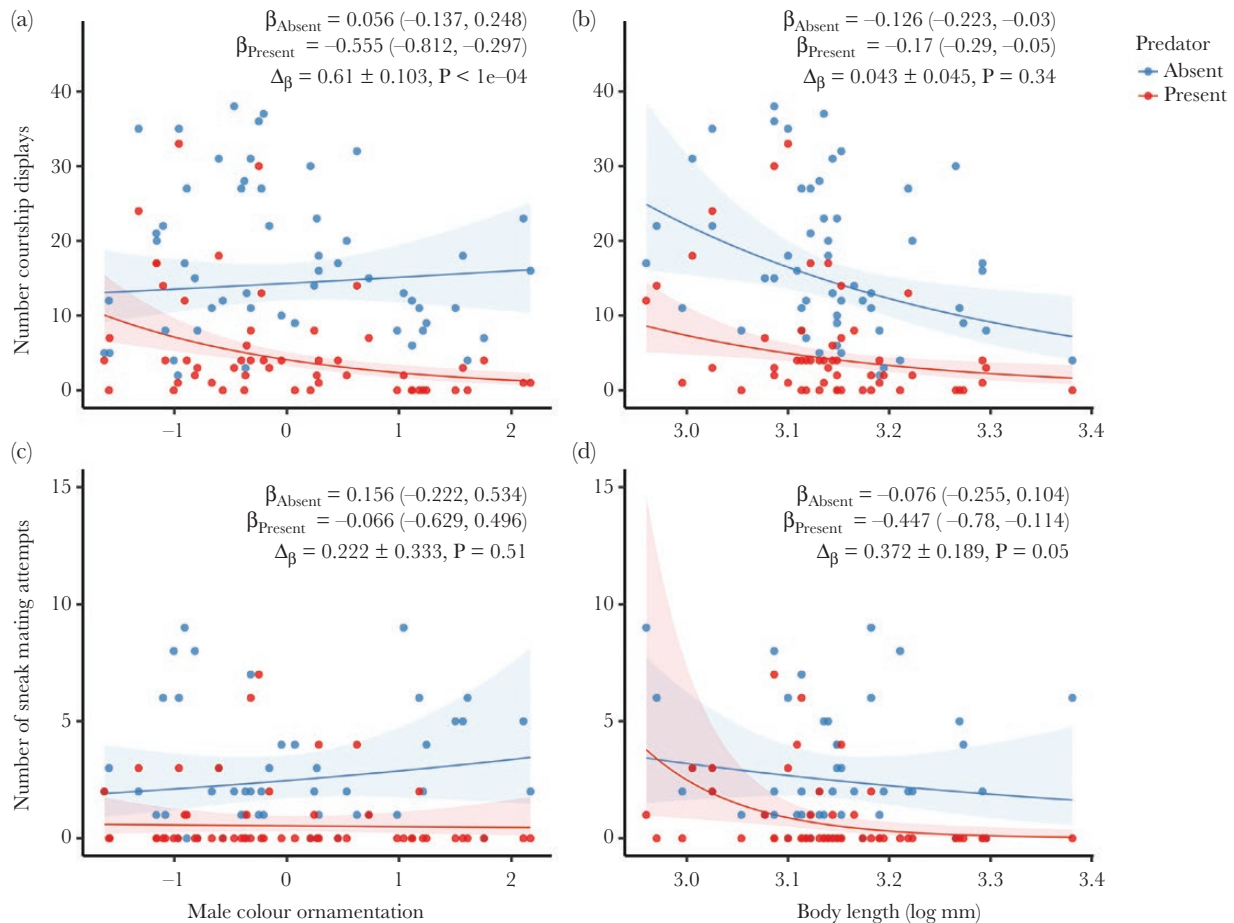


Figure 2

Box plots of mating behavior frequencies exhibited by Quaré River male guppies ($N = 53$) in both the absence and presence of a cichlid fish predator, separately for each mating tactic (courtship displays and sneak mating attempts). Box plots show the median, 25% and 75% inter-quartiles, and the 5% and 95% range of values (whiskers) of the observed behavioral responses. Raw data are shown as small points. Larger points with error bars denote predicted mean values and associated 95% CIs from a zero-inflated Poisson model (Table 1).

**Figure 3**

Relationships between the frequencies (acts/15 min) of courtship displays (panels a and b) and sneak mating attempts (panels c and d) expressed by individual male guppies in both the absence (blue lines and data points) and presence (red lines and data points) of a fish predator and, respectively, their body color ornamentation (proportion of total color, mean-centered and scaled to standard deviation units) and body length (mm, log). Trend lines show each of the predicted relationships from respective GLMMs (i.e., zero-inflated Poisson for courtship displays, Table 2; negative binomial for sneak mating attempts, Table 3), holding other variables constant. The colored ribbons indicate 95% CI, and points are the raw data. Inset within each panel shows the slope (β) and associated 95% CI of each regression trend line, along with a test of their difference ($\Delta\beta$). Note that for courtship displays the β estimates and contrasts are based on the Poisson part of the model only.

minimize any stress in the guppies and the number of fish used in the current study (see [Supplementary Material](#), Animal Ethical Statement for more details).

RESULTS

Of the two alternative mating tactics, courtship displaying toward females was the one predominantly employed by male guppies in both predator treatments (Figure 2), with $85.4 \pm 2.1\%$ (mean $\pm$ SE) and $87.3 \pm 3.3\%$ of their total mating effort consisting of courtship displays in the predator-absent and predator-present treatments, respectively. Overall, the frequencies of each of the two mating tactics exhibited by focal males differed significantly within the paired predator treatments (Figure 2). As predicted a priori, males significantly suppressed the expression of both their courtship displays and sneak mating attempts in response to an increased apparent threat of predation, with all other variables held constant (Figure 2, Table 1). Our omnibus zero-inflated Poisson model showed no significant interaction effect between mating tactic type and predator treatment on the likelihood of zeroes (including this interaction in

the zero-inflated component of the model caused some convergence issues and did not improve model fit, causing AIC to increase by 0.72). We did find significant main effects on the likelihood of zeroes, such that the predator-present treatment and the sneak mating tactic type both increased the likelihood of zeroes (Table 1). In the count component of the model, we found a significant interaction between mating tactic type and predator treatment on the count frequency; sneak mating attempts were overall significantly lower in frequency than courtship displays, and the negative effect of the predator-present treatment on count frequency was significantly less pronounced for sneak attempts relative to courtship displays (Figure 2, Table 1). Both alternative mating tactics expressed by individual male guppies were therefore predator threat sensitive (cf. Figure 1B), with the courtship tactic being significantly more so than the coercive sneak tactic as expected theoretically.

Our separate zero-inflated Poisson GLMM revealed that male courtship display rate was significantly negatively influenced by predator treatment, male body length and male color ornamentation contingent on predator treatment (Table 2). Although courtship display rate did not vary with male color ornamentation in

Table 1

Parameter estimates of an omnibus zero-inflated Poisson GLMM testing for the main fixed effects of predator treatment and mating tactic type, and their interaction, on the rate of mating behaviors (jointly courtship displays and sneak mating attempts) exhibited by male Trinidadian guppies, with predator treatment order (fixed effect) and the identity of individual subjects (random effect) controlled for

			Mating behavior rate				
	Effect	Predictors	Log-mean	SE	CI	P	ΔAIC
Count model	Fixed	Intercept	2.67	0.09	2.49, 2.85	<0.001	7.6
		Predator treatment (present)	−0.93	0.07	−1.07, −0.80	<0.001	
		Mating tactic type (sneak)	−1.69	0.16	−2.00, −1.39	<0.001	
		Treatment order	0.02	0.14	−0.26, 0.29	0.913	
		Predator trt × Mating tactic type	0.76	0.23	0.30, 1.22	0.001	
	Random	σ ² _{MaleID.Courtship}	0.36				
		σ ² _{MaleID.Sneak}	0.70				
Zero-inflated model	Fixed	γ _{MaleID.Courtship, MaleID.Sneak}	−0.78				
		Intercept	−5.83	1.93	−9.61, −2.04	0.003	
		Predator treatment (present)	3.97	1.31	1.40, 6.53	0.002	
		Mating tactic type (sneak)	3.14	1.18	0.83, 5.44	0.008	
	Random	σ ² _{MaleID}	2.73				

Significant terms are boldfaced. Δ AIC shows the difference in AIC for models including or excluding the interaction term.

Table 2

Parameter estimates of a zero-inflated Poisson GLMM testing for the main fixed effects of predator treatment, body color ornamentation (proportion of body colored) and body length (log mm), and the interaction between predator treatment and color ornamentation, on the rate of courtship displays exhibited by male Trinidadian guppies, with predator treatment order (fixed effect) and the identity of individual subjects (random effect) controlled for

				Courtship display rate			
	Effect	Predictors	Log-mean	SE	CI	P	ΔAIC
Count model	Fixed	Intercept	2.67	0.09	2.50, 2.84	<0.001	47.5
		Predator treatment (present)	−1.22	0.08	−1.39, −1.06	<0.001	
		Color ornamentation	0.06	0.10	−0.13, 0.25	0.517	
		Body length	−0.26	0.09	−0.44, −0.08	0.005	
		Treatment order	0.25	0.18	−0.09, 0.60	0.153	
		Predator trt × Color ornamentation	−0.65	0.09	−0.83, −0.46	<0.001	
	Random	σ ² _{MaleID}	0.32				
Zero-inflated model	Fixed	Intercept	−2.77	0.51	−3.76, −1.78	<0.001	
	Random	σ ² _{MaleID}	0.0				

Significant terms are boldfaced. The remainder of the caption is as for Table 1.



the absence of the predator, it was overall significantly lower and negatively correlated with male color ornamentation in the presence of the predator (estimate $\pm$ SE = -0.65 ± 0.09 , $\chi^2_1 = 49.5$, $P < 0.001$; Δ AIC = 47.5; Table 2, Figure 3A). Therefore, the degree of predator-threat sensitivity of courtship behavior (significant Δ β ; Figure 3A) is strongly predicted by male color ornamentation, with more color ornamented individuals being relatively more risk averse and threat sensitive in courting females than less ornamented conspecifics when all other variables in the model are controlled for, as predicted a priori (cf. Figure 1C). In comparison, courtship display rate significantly decreased with increasing male body length in both predator treatments when all other variables in the model are controlled for; that is, larger males courted at lower frequencies than smaller males irrespective of predator treatment (Table 2, Figure 3B). Because there was no significant statistical interaction between predation treatment and male body length (estimate $\pm$ SE = -0.08 ± 0.09 , $\chi^2_1 = 0.92$, $P = 0.34$; Δ AIC = 1.08) and thus Δ β is not significant (Figure 3B), we found no evidence for male

body length predicting the degree of threat sensitivity in courtship behavior.

The negative binomial GLMM showed that the frequency of male sneak mating attempts was significantly negatively influenced by predator treatment (Table 3; Figure 3C and D). We found no effect of the interaction between predator treatment and male color ornamentation (estimate $\pm$ SE = -0.22 ± 0.33 , $\chi^2_1 = 0.44$, $P = 0.51$; Δ AIC = 1.56; Δ β nonsignificant), nor a main effect of color ornamentation (Figure 3C; Table 3). However, there was a significant negative interaction between predator treatment and male body length (estimate $\pm$ SE = -0.82 ± 0.34 , $\chi^2_1 = 6.59$, $P = 0.018$; Δ AIC = 4.59; Δ β significant), but no main effect of body length (Figure 3D; Table 3), such that the frequency of sneak mating attempts is relatively invariant with male body length in the absence of the predator but significantly negatively correlated with male body length in the presence of the predator (Figure 3D). These results indicate that males reduced their coercive sneak mating attempts in the presence of the predator, and suggest that male body

Table 3

Parameter estimates of a negative binomial GLMM testing for the main fixed effects of predator treatment, body color ornamentation (proportion of body colored) and body length (log mm), and the interaction between predator treatment and body length, on the rate of sneak mating attempts exhibited by male Trinidadian guppies, with predator treatment order (fixed effect) and the identity of individual subjects (random effect) controlled for

				Sneak mating attempt rate			
	Effect	Predictors	Log-mean	SE	CI	P	ΔAIC
Count model	Fixed	Intercept	0.92	0.18	0.56, 1.28	<0.001	4.6
		Predator treatment (present)	−1.49	0.32	−2.12, −0.87	<0.001	
		Color ornamentation	0.09	0.16	−0.23, 0.41	0.583	
		Body length	−0.12	0.17	−0.45, 0.22	0.498	
		Treatment order	−0.00	0.29	−0.57, 0.56	0.994	
		Predator trt × Body length	−0.82	0.34	−1.49, −0.14	0.018	
	Random	σ ² _{MaleID}	0.06				

Significant terms are boldfaced. The remainder of the caption is as for Table 1.

color ornamentation did not covary overall with their sneak mating behavior and did not predict its threat sensitivity (Figure 3C), whereas male body length predicts (albeit modestly) the degree of threat sensitivity in sneak mating behavior, with larger males being relatively more risk averse and threat sensitive in expressing sneak mating behavior than smaller males (Figure 3D), as expected a priori (cf. Figure 1C).

Overall, based on our results, we can reject the null hypothesis of no influence of predation risk perception, male body coloration and male body length on the threat sensitivity of either male mating tactics (Figure 1A). Alternatively, our current findings lend support to the hypotheses that both courtship displays and coercive (sneak) mating attempts in Trinidadian male guppies are predator threat sensitive (Figure 1B) and that the degree of threat sensitivity of both these mating tactics is phenotype dependent (Figure 1C).

DISCUSSION

In this study, we tested the hypothesis, based on theory (Helfman 1989; Clark 1994; Engqvist et al. 2015), that more color ornamented, larger and/or vigorously courting males should be more threat sensitive and flexibly suppress their mating effort relatively more than their less color ornamented, smaller and/or less vigorously courting counterparts under predation hazard, because the former are more likely to incur higher predation costs when searching for, attracting and mating with females (e.g., Magnhagen 1991; Zuk and Kolluru 1998; White et al. 2022) and have more to lose in terms of “assets” according to the asset-protection principle (Clark 1994) than the latter, all else being equal. We found that individual male Trinidadian guppies suppressed their overall mating effort in response to a perceived predation threat, decreasing the frequency of their (presumably more conspicuous) courtship displays significantly more on average than the frequency of their sneak mating behavior. The degree of threat-sensitivity in the expression of these two alternative mating tactics was dependent on the phenotypic attractiveness (color ornamentation, body size) of individual males.

Male guppies significantly decreased the frequencies of both their courtship displays and sneak mating attempts directed toward females in response to a staged immediate threat of fish predation, as predicted a priori (Figure 1B). On average, the observed decrease in their courtship displays was significantly greater than the decrease in their sneak mating attempts under elevated predation

threat perception. Both alternative mating tactics expressed by individual male guppies were therefore predator threat sensitive, with the courtship tactic being significantly more so overall than the coercive sneak mating tactic as expected theoretically. This is a relatively novel finding. Assuming that high mating effort (i.e., high rates of courtship displaying and sneak mating attempts) elevates the performer’s risk of predation (Lima and Dill 1990; Magnhagen 1991; Sih 1994; Zuk and Kolluru 1998; Clark et al. 2016; Johnson and Candolin 2017; White et al. 2022), the above results are consistent with the threat-sensitive hypothesis (Helfman 1989) and presumably reflect adaptive behavioral plasticity, in that such threat sensitivity in mating behavior would mitigate a male’s immediate risk of predation (Lerch and Servadio 2023). We consider below alternative interpretations of our main findings.

Previous studies have reported that, within Trinidadian guppy populations, males typically suppress their courtship display rate (Endler 1987; Magurran and Seghers 1990; Magurran and Nowak 1991; Reynolds 1993; Reynolds et al. 1993; Godin 1995; Godin and Briggs 1996; Glavaschi et al. 2020, 2022) and conversely either increase (Endler 1987; Magurran and Seghers 1990; Magurran and Nowak 1991; Reynolds 1993; Godin 1995; Evans et al. 2002; Glavaschi et al. 2020, 2022) or do not change (Reynolds et al. 1993; Godin and Briggs 1996; Kelly and Godin 2001) their rate or proportion of sneak mating attempts in response to an acute apparent increase in their ambient risk of predation. Threat sensitivity in male mating behavior appears to be selected for and has evolved in a graded manner that reflects among-population variation in predation intensity experienced by wild guppies in Trinidad. Male guppies in populations that have evolved under high fish predation intensity tend on average to court females less often and attempt sneak copulations more frequently than males in populations with a lower average risk of predation (Luyten and Liley 1985; Endler 1987; Magurran and Seghers 1994; Godin 1995; Evans and Magurran 1999). Our aforementioned results corroborate those of the latter studies in that wild-caught male guppies from the high-predation Quaré River suppressed their courtship display rate under a perceived predation threat, but differ from these previous studies in that our male subjects also concurrently suppressed their sneak mating attempts under predation threat, as was previously reported by Dill et al. (1999) for Quaré River males. Therefore, in our current study and that of Dill et al. (1999), Quaré River male guppies did not compensate for a suppressed courtship display rate with an increased rate or proportion of sneak

mating attempts when exposed to a live fish predator nearby, as other studies have previously reported for male Trinidadian guppies (Endler 1987; Magurran and Seghers 1990; Magurran and Nowak 1991; Reynolds 1993; Godin 1995; Evans et al. 2002; Glavaschi et al. 2020, 2022). This difference in our results and those of the latter studies might be attributable to differences in the fish's assessment of the nature of the predation threat, ecological and demographic differences between study populations, differences in the internal state of individual males and differences in female reproductive state and behavior between studies, differences in experimental protocols, and whether the studies were carried out in the field or laboratory or on known individuals or at the population level, for example. Additionally, changes in male mating behavior in the presence of predators could have been driven indirectly by predator-mediated changes in female behavior in some studies (Magurran and Nowak 1991; Dill et al. 1999; Evans et al. 2002) but not in others. Males of other species have also been found to similarly decrease the expression of their conspicuous courtship displays when under increased predation risk perception (reviewed by Lima and Dill 1990; Sih 1994; Zuk and Kolluru 1998; Bernal and Page 2023), but there is generally a paucity of information on whether and how alternative male mating tactics, such as satellite or sneak coercive mating, are modified by individual males in response to perceived immediate threat of predation.

Statistically controlling for body length, we showed that more color-ornamented male guppies were relatively more threat sensitive in their courtship displays, but not in their sneak mating attempts, under predation hazard than drabber males were, as expected theoretically (Figure 1C). Controlling for body coloration, larger males tended to exhibit lower courtship and sneak mating efforts than smaller males did in both predation treatments, but body length influenced the degree of threat sensitivity only for sneak mating behavior. Larger males were relatively more threat sensitive in their coercive mating attempts than smaller males. These results are the most novel ones of our current study and are largely consistent with both the threat-sensitive hypothesis (Helfman 1989) and asset-protection principle (Clark 1994), assuming that more ornamented and larger male guppies are at greater risk of mortality to predation and thus have more to lose in terms of assets than less ornamented and smaller males in the face of predation hazard (see supporting evidence for this assumption in the Methods section, subsection Model study system).

Proximally, different alternative mechanisms could underlie the suppression of male mating tactics under an apparent immediate threat of predation as observed in our current study. First, individual wild-caught male Trinidadian guppies perceived and responded directly to an immediate threat of predation (i.e., the presence of a predatory fish nearby) by suppressing the frequencies of their courtship displays and sneak mating attempts, conditional on their individual body coloration and body length, so as to render themselves less conspicuous and vulnerable to predation, in a manner consistent with the threat-sensitive (Helfman 1989) and asset-protection (Clark 1994) hypotheses. These wild-caught male guppies would have previously encountered fish predators and female guppies in the Quaré River and likely learned through these experiences/encounters about their relative attractiveness to both females and predators based on their color ornamentation and body size. Second, males could alternatively have responded indirectly to a perceived predation threat by reducing their courtship display rate because females are less responsive to them owing to their increased engagement in antipredator behavior (Magurran

and Nowak 1991; Evans et al. 2002). If so, then we should have expected males to take advantage of female inattentiveness to them by increasing their rate of coercive sneak mating attempts (Magurran and Nowak 1991; Evans et al. 2002), which we did not observe in the current study. Third, more color ornamented and/or larger males might proportionately receive less sexual attention from females because they attract predators and consequently they court less under predation risk than less ornamented males (Dill et al. 1999). The latter alternative mechanism was not likely operating here because we used gravid females, who are generally not sexually receptive to males (Liley 1966; Houde 1997), as stimulus fish in both predation treatments in the current study. There is thus no a priori reason to expect that unreceptive female guppies would direct any less attention to more color ornamented or larger males over less ornamented and smaller ones when the predator was present compared to when it was absent.

Of the above three alternative mechanisms, we propose that the first one (i.e., direct male behavioral response to a threat of predation) is the most likely proximate explanation for our main findings and conclusion of direct, phenotype-dependent threat sensitivity in the mating tactics of male guppies. However, our current study was not specifically designed to experimentally differentiate between these alternative, non-exclusive proximate mechanisms underlying our results. Two previous studies on the guppy lend support to the second and third proximate mechanisms listed above. Evans et al. (2002) experimentally uncoupled the direct effects of predation threat on male guppy mating behavior from those potentially attributable to predator-mediated changes in female antipredator behavior, and reported that the observed switch from predominantly expressing courtship displays to coercive sneak matings by male guppies in apparent high-risk situations was indirectly mediated by adjustments in female behavior rather than directly by the predator. Dill et al. (1999) experimentally manipulated whether either male or female guppies separately, or both sexes concurrently, had access to visual stimuli from a nearby fish predator. Compared to a control treatment (no predation threat), they found that when only females were provided information about an apparent predation threat, males more strongly reduced their mating effort (both courtship displays and sneak matings) than in the other treatments. This result suggests that female perception of danger and associated changes in their behavior (including potentially reduced sexual receptivity to male mating effort) indirectly mediated the observed suppression of the mating effort of males under increased predation risk.

Our findings contribute importantly to a better understanding of the relationship between the degree of threat sensitivity in sexual signaling and mating effort expressed by reproductively active males and their sexual attractiveness to females, as measured by their color ornamentation and body size, for example. Hitherto, only a few studies on other species had shown that individual males with greater expression of sexually selected traits similarly tend to be more cautious and risk averse in their sexual signaling and mating behavior (e.g., Candolin 1998; Hedrick 2000; Kraak et al. 2000; Fowler-Finn and Hebets 2011; McQueen et al. 2017; but see Cordes et al. 2014 for an exception).

In conclusion, our current study highlights the apparent adaptive plasticity of the alternative mating tactics of individual male Trinidadian guppies, conditional on their sexual attractiveness, under varying perceived threat of predation. We provided experimental evidence strongly suggesting that both alternative male mating tactics in this species are plastic, predation threat sensitive and phenotype dependent, in that the degree of threat sensitivity of

courtship and sneak mating behavior depended on individual body coloration and body length, two sexually selected male traits in the guppy that are also potentially attractive to visually guided predators. Concurrently suppressing both courtship display and sneak mating rates when faced with a perceived immediate threat of predation would likely mitigate an individual male guppy's risk of mortality to predation in the short term, but at the potential costs of reduced sexual attractiveness (Godin and Briggs 1996; Kodric-Brown and Nicoletto 2001; Kolluru et al. 2009), mating opportunities (Evans and Magurran 2001; Kelly and Godin 2001; Devigili et al. 2015) and reproductive success (Glavaschi et al. 2020, 2022).

At the population level, such phenotype-dependent threat sensitivity in alternative mating tactics among males may contribute to the maintenance of male color polymorphism, body size variation and variation in the frequencies of courtship displays and sneak mating in this species under varying predation risk (cf. Lerch and Servedio 2023). Moreover, non-consummatory effects of immediate predation risk on male mating tactics as reported in our current study, along with concurrent potential predator-induced reductions in female mating rate, responsiveness to male courtship and preference for conspicuous male traits (Magurran and Nowak 1991; Godin and Briggs 1996; Dill et al. 1999; Evans et al. 2002; Glavaschi et al. 2020, 2022), have important implications for the opportunity for sexual selection and polyandry in the Trinidadian guppy (Glavaschi et al. 2020, 2022). The latter predator-mediated effects on the direction, strength and shape of realized multivariate sexual selection on male pre-copulatory and post-copulatory traits in the guppy are complex and generally nonlinear (i.e., disruptive), resulting in different male phenotypes achieving similar high reproductive fitness and thus contributing to the maintenance of male color polymorphism and variation in mating tactics within populations (Glavaschi et al. 2020, 2022). Given the empirical results of Devigili et al. (2015) and Glavaschi et al. (2020, 2022), the theoretical results of Lerch and Servedio (2023) and our current findings, it would therefore be interesting to investigate in future whether the magnitude of non-consummatory predator-mediated effects on sexual selection depends on the particular distribution of sexually attractive traits among males (e.g., Auge et al. 2016) and on the degree of heterogeneity in the ambient operational sex ratio and the relative frequencies of alternative male mating tactics (e.g., Barbosa et al. 2018) within populations of the Trinidadian guppy and as well as in other species.

SUPPLEMENTARY MATERIAL

Supplementary material can be found at <http://www.behco.oxfordjournals.org/>

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AUTHOR CONTRIBUTIONS

Jean-Guy Godin (Conceptualization [Lead], Data curation [Equal], Formal analysis [Equal], Funding acquisition [Lead], Investigation [Lead],

Methodology [Lead], Project administration [Lead], Resources [Lead], Software [Supporting], Supervision [Lead], Validation [Equal], Visualization [Supporting], Writing – original draft [Lead], Writing – review & editing [Lead]), Heather McDonough (Investigation [Supporting], Methodology [Supporting]), and Thomas Houslay (Data curation [Equal], Formal analysis [Equal], Methodology [Supporting], Software [Lead], Validation [Equal], Visualization [Lead], Writing – review & editing [Supporting])

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY

Analyses reported in this article can be reproduced using the data provided by Godin et al. (2023).

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